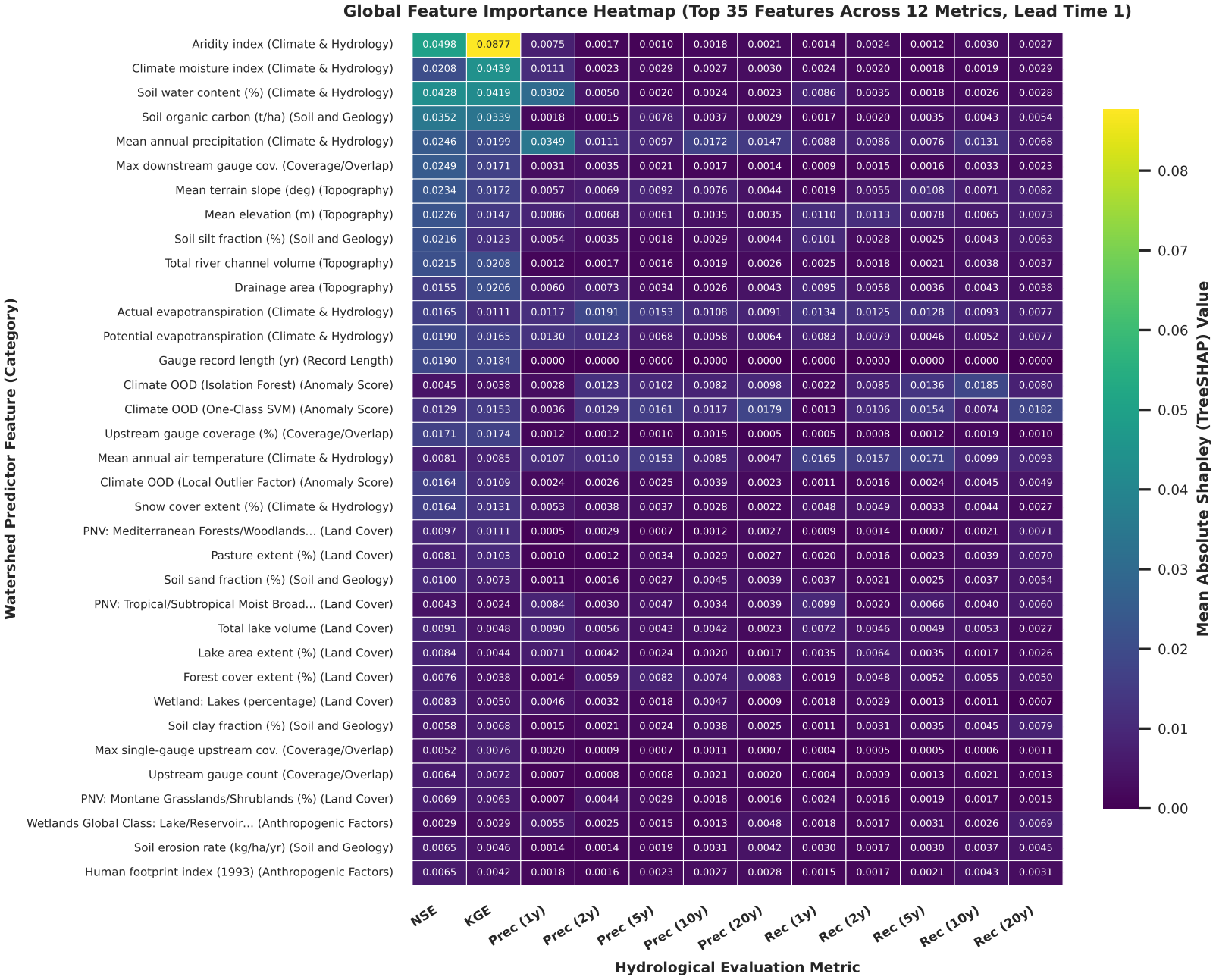
Global Feature Importance Heatmap (Top 35 Features Across 12 Metrics, Lead Time 1)
Watershed Predictor Feature (Category)
Aridity index (Climate & Hydrology) 0.0498 0.0877 0.0075 0.0017 0.0010 0.0018 0.0021 0.0014 0.0024 0.0012 0.0030 0.0027
Climate moisture index (Climate & Hydrology) 0.0208 0.0439 0.0111 0.0023 0.0029 0.0027 0.0030 0.0024 0.0020 0.0018 0.0019 0.0029
Soil water content (%) (Climate & Hydrology) 0.0428 0.0419 0.0302 0.0050 0.0020 0.0024 0.0023 0.0086 0.0035 0.0018 0.0026 0.0028
Soil organic carbon (t/ha) (Soil and Geology) 0.0352 0.0339 0.0018 0.0015 0.0078 0.0037 0.0029 0.0017 0.0020 0.0035 0.0043 0.0054
Mean annual precipitation (Climate & Hydrology) 0.0246 0.0199 0.0349 0.0111 0.0097 0.0172 0.0147 0.0088 0.0086 0.0076 0.0131 0.0068
Max downstream gauge cov. (Coverage/Overlap) 0.0249 0.0171 0.0031 0.0035 0.0021 0.0017 0.0014 0.0009 0.0015 0.0016 0.0033 0.0023
Mean terrain slope (deg) (Topography) 0.0234 0.0172 0.0057 0.0069 0.0092 0.0076 0.0044 0.0019 0.0055 0.0108 0.0071 0.0082
Mean elevation (m) (Topography) 0.0226 0.0147 0.0086 0.0068 0.0061 0.0035 0.0035 0.0110 0.0113 0.0078 0.0065 0.0073
Soil silt fraction (%) (Soil and Geology) 0.0216 0.0123 0.0054 0.0035 0.0018 0.0029 0.0044 0.0101 0.0028 0.0025 0.0043 0.0063
Total river channel volume (Topography) 0.0215 0.0208 0.0012 0.0017 0.0016 0.0019 0.0026 0.0025 0.0018 0.0021 0.0038 0.0037
Drainage area (Topography) 0.0155 0.0206 0.0060 0.0073 0.0034 0.0026 0.0043 0.0095 0.0058 0.0036 0.0043 0.0038
Actual evapotranspiration (Climate & Hydrology) 0.0165 0.0111 0.0117 0.0191 0.0153 0.0108 0.0091 0.0134 0.0125 0.0128 0.0093 0.0077
Potential evapotranspiration (Climate & Hydrology) 0.0190 0.0165 0.0130 0.0123 0.0068 0.0058 0.0064 0.0083 0.0079 0.0046 0.0052 0.0077
Gauge record length (yr) (Record Length) 0.0190 0.0184 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000
Climate OOD (Isolation Forest) (Anomaly Score) 0.0045 0.0038 0.0028 0.0123 0.0102 0.0082 0.0098 0.0022 0.0085 0.0136 0.0185 0.0080
Climate OOD (One-Class SVM) (Anomaly Score) 0.0129 0.0153 0.0036 0.0129 0.0161 0.0117 0.0179 0.0013 0.0106 0.0154 0.0074 0.0182
Upstream gauge coverage (%) (Coverage/Overlap) 0.0171 0.0174 0.0012 0.0012 0.0010 0.0015 0.0005 0.0005 0.0008 0.0012 0.0019 0.0010
Mean annual air temperature (Climate & Hydrology) 0.0081 0.0085 0.0107 0.0110 0.0153 0.0085 0.0047 0.0165 0.0157 0.0171 0.0099 0.0093
Climate OOD (Local Outlier Factor) (Anomaly Score) 0.0164 0.0109 0.0024 0.0026 0.0025 0.0039 0.0023 0.0011 0.0016 0.0024 0.0045 0.0049
Snow cover extent (%) (Climate & Hydrology) 0.0164 0.0131 0.0053 0.0038 0.0037 0.0028 0.0022 0.0048 0.0049 0.0033 0.0044 0.0027
PNV: Mediterranean Forests/Woodlands... (Land Cover) 0.0097 0.0111 0.0005 0.0029 0.0007 0.0012 0.0027 0.0009 0.0014 0.0007 0.0021 0.0071
Pasture extent (%) (Land Cover) 0.0081 0.0103 0.0010 0.0012 0.0034 0.0029 0.0027 0.0020 0.0016 0.0023 0.0039 0.0070
Soil sand fraction (%) (Soil and Geology) 0.0100 0.0073 0.0011 0.0016 0.0027 0.0045 0.0039 0.0037 0.0021 0.0025 0.0037 0.0054
PNV: Tropical/Subtropical Moist Broad... (Land Cover) 0.0043 0.0024 0.0084 0.0030 0.0047 0.0034 0.0039 0.0099 0.0020 0.0066 0.0040 0.0060
Total lake volume (Land Cover) 0.0091 0.0048 0.0090 0.0056 0.0043 0.0042 0.0023 0.0072 0.0046 0.0049 0.0053 0.0027
Lake area extent (%) (Land Cover) 0.0084 0.0044 0.0071 0.0042 0.0024 0.0020 0.0017 0.0035 0.0064 0.0035 0.0017 0.0026
Forest cover extent (%) (Land Cover) 0.0076 0.0038 0.0014 0.0059 0.0082 0.0074 0.0083 0.0019 0.0048 0.0052 0.0055 0.0050
Wetland: Lakes (percentage) (Land Cover) 0.0083 0.0050 0.0046 0.0032 0.0018 0.0047 0.0009 0.0018 0.0029 0.0013 0.0011 0.0007
Soil clay fraction (%) (Soil and Geology) 0.0058 0.0068 0.0015 0.0021 0.0024 0.0038 0.0025 0.0011 0.0031 0.0035 0.0045 0.0079
Max single-gauge upstream cov. (Coverage/Overlap) 0.0052 0.0076 0.0020 0.0009 0.0007 0.0011 0.0007 0.0004 0.0005 0.0005 0.0006 0.0011
Upstream gauge count (Coverage/Overlap) 0.0064 0.0072 0.0007 0.0008 0.0008 0.0021 0.0020 0.0004 0.0009 0.0013 0.0021 0.0013
PNV: Montane Grasslands/Shrublands (%) (Land Cover) 0.0069 0.0063 0.0007 0.0044 0.0029 0.0018 0.0016 0.0024 0.0016 0.0019 0.0017 0.0015
Wetlands Global Class: Lake/Reservoir... (Anthropogenic Factors) 0.0029 0.0029 0.0055 0.0025 0.0015 0.0013 0.0048 0.0018 0.0017 0.0031 0.0026 0.0069
Soil erosion rate (kg/ha/yr) (Soil and Geology) 0.0065 0.0046 0.0014 0.0014 0.0019 0.0031 0.0042 0.0030 0.0017 0.0030 0.0037 0.0045
Human footprint index (1993) (Anthropogenic Factors) 0.0065 0.0042 0.0018 0.0016 0.0023 0.0027 0.0028 0.0015 0.0017 0.0021 0.0043 0.0031
NSE
KGE
Prec (1y)
Prec (2y)
Prec (5y)
Prec (10y)
Prec (20y)
Rec (1y)
Rec (2y)
Rec (5y)
Rec (10y)
Rec (20y)
Hydrological Evaluation Metric
Mean Absolute Shapley (TreeSHAP) Value
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08